

15

FEATURE

FEATURE 15 OF 17 - WHAT THE MONSOON DOES TO A SATELLITE MODEL

Cloud Dropout

An experiment that set out to test robustness to cloud and instead found a warm bias in the deliverable.

PORT	PAGE FILE	SCRIPT	COST
8515	app/phase2/dropout_page.py	scripts/phase2/run_cloud_dropout.py	about 31 s, no retraining

Our model reads sea-surface temperature as its first channel, and during the monsoon thick cloud blinds infrared temperature retrieval for days at a time over large parts of this basin. So the question is not whether the model degrades under cloud. It is how fast - and the answer turned out to be about something else entirely.

1. What this feature is, in plain words

The experiment is simple to describe. Take the shipped model, unchanged. Blank out a fraction of the sea-surface temperature pixels in the **test** input, as cloud would. Score it. Repeat from 0% blanked to 100%, three different random masks at each level, and draw the curve.

One detail decides whether the whole experiment means anything. The blanking is done in **real physical units, on the raw data, before the model's normalisation step**. Do it the other way round - blank after normalisation, or blank to zero - and every 'missing' pixel arrives at the network as zero, which after normalisation *is the channel average*: a perfectly plausible ordinary-temperature pixel. The error would barely move and the result would read '*robust to 60% cloud cover*' having tested precisely nothing.

And then the curve did something nobody expected. **Blanking 15% of the sea-surface temperature made the model better** - 0.8950 against the control's 0.9078, with a spread across random masks of 0.0002, so sixty times the noise. It would have been easy, and completely wrong, to report that as tolerance of cloud.

It is two errors partially cancelling. The shipped model runs **0.1003 degC warm**. A blanked pixel reaches the network as the channel average, which pulls the prediction cooler. The best score sits essentially where the bias crosses zero, at about 19%. So the finding is not about cloud at all - **it is about the deliverable, and it says a bias correction is worth about 0.013 degC, free**.

THE EVERYDAY PICTURE

A student who consistently over-estimates every answer will score better on a test where some questions are removed at random - not because removing questions helps, but because it accidentally cancels a habit. Reporting that as 'robust to missing questions' would be the wrong lesson entirely.

2. What you actually see on the screen

One curve, and the reasoning that stops it being misread:

- Error against fraction of sea-surface temperature blanked, from 0% to 100%, averaged over three random masks each.
- The bias curve on the same axis - which is what makes the dip explicable rather than mysterious.
- The control point, which reproduces the shipped model's recorded score **exactly**.
- A written statement that the dip is a bias cancellation and **not** robustness to cloud.

- A stated caveat that the masking is random, while real cloud is spatially clustered and persistent - so a genuine overcast should hurt more than this curve shows.

3. How to run it, step by step

This page renders a stored result and runs no inference itself, because a results page that recomputed its own numbers would be a second definition of them. To regenerate the result:

RUN THIS

```
PYTHONPATH=src python scripts/phase2/run_cloud_dropout.py --tag sat_7ch_s42 --daily-dir data/processed/daily_sat/v001
```

- 1 Run the experiment script above. It takes about 31 seconds for the full ten-point sweep and requires no retraining.
- 2 Note the control check in the output - the script **refuses to write its result file** if the 0% point does not reproduce the shipped model's recorded score.
- 3 Launch the page with the second command below and open **http://localhost:8515**.
- 4 Show the curve and let the audience see the dip at 15%.
- 5 Ask them what it means, then give the real answer: it is a bias cancellation, and it is a finding about our model rather than about clouds.
- 6 Finish on the caveat: real cloud is clustered, not random, so a real overcast should be worse than this.

OPEN THE RESULTS PAGE

```
PYTHONPATH=src python -m streamlit run app/phase2/dropout_page.py --server.port 8515
```

THE CONTROL IS THE WHOLE EXPERIMENT

At 0% blanked the harness must reproduce the checkpoint's own recorded score, and it does - both read 0.9077608087441584. If it did not, every degradation measured below it would be harness error rather than weather, so the script refuses to write anything at all.

4. Why the problem statement needs this

The problem statement asks for a satellite-input system over a basin dominated by the monsoon. Cloud is the single most obvious operational failure mode for such a system, and not testing it would leave the most likely first question from an operational reviewer unanswered.

It also produced an actionable finding about the deliverable that no other experiment had surfaced this cleanly: the model runs warm by a measurable amount, and removing that is worth about 0.013 degC.

PS requirement	How this feature answers it
Robustness of a satellite-input system	Quantifies degradation from 0% to 100% missing sea-surface temperature, with three mask draws per point.
Honest evaluation	The control reproduces the shipped score exactly, and the script refuses to write otherwise.
Report bias	Traced the warm bias through an independent route and quantified what correcting it would be worth.

5. The measured numbers behind it

The full sweep, on 962 Argo profiles with the shipped checkpoint unmodified [VERIFIED]:

SST blanked	0%	10%	15%	20%	30%	50%	70%	100%
RMSE (degC)	0.9078	0.8957	0.8950	0.8973	0.9118	0.9857	1.1087	1.4114
Bias (degC)	+0.100	+0.047	+0.021	-0.005	-0.059	-0.163	-0.267	-0.471

- **Read the two rows together.** The best error is at 15% and the bias crosses zero at about 19%. That is not a coincidence - it is the mechanism.
- Past the minimum the degradation is real and monotone: **+0.50 degC by 100% blanked**, with skill against climatology going **negative at 90%**. Under total blindness the model is worse than simply quoting the long-term average, which is the correct and reassuring behaviour.
- **The bias correction was NOT applied.** That is a change to a frozen model and belongs to the person who owns it - so it is recorded as a decision somebody takes rather than something nobody noticed.

6. The impact - and who it is for

Who	What changes for them
Operational users	A quantified answer to the first question anyone will ask about a monsoon-season satellite product.
The team	An independent confirmation of the warm bias, and a price tag on fixing it.
The jury	A demonstration of what good experimental design looks like - including the mistake that would have made it meaningless, stated openly.
Future model work	A named architectural gap: the encoder has no way to know a pixel is missing.

WHY SOCIETY GAINS FROM THIS

A monsoon-season ocean product that has never been tested under monsoon-season cloud is a product that will fail exactly when it is needed most. Measuring the failure mode before deployment - and reporting that a convenient-looking result was actually something else - is what separates an operational tool from a demonstration.

7. Say this to the jury

We blanked out sea-surface temperature the way monsoon cloud does, and the curve did something surprising: at 15% blanked the model got better. It would have been very easy to report that as tolerance of cloud cover. It is not. Our model runs 0.1 degC warm, a blanked pixel arrives at the network as the channel average which pulls it cooler, and the best score sits exactly where the bias crosses zero. So this experiment did not find robustness - it found a warm bias in our own deliverable, and told us that correcting it is worth about 0.013 degC.

This is the best story in the pack about how the team thinks. Tell it in full.

8. Questions the jury will ask

Q. So your model is robust to cloud cover?

A. No, and we are careful about that. The improvement at light masking is an accident of error cancellation. Past about 20% the degradation is real and monotone - half a degree by total blindness, with skill going negative at 90%. And the real caveat is worse: our masking is random, while real cloud is clustered and persistent, so a genuine overcast should hurt more than this curve shows.

Q. How do you know the improvement is not just noise?

A. The spread across three independent random masks is 0.0002 degC, and the improvement is 0.0128 - sixty times the noise floor. And the control at 0% reproduces the shipped model's recorded score to fifteen decimal places, so the harness itself is verified.

Q. Why not simply apply the bias correction and take the better number?

A. Because that is a change to a frozen, published model, and it belongs to the person who owns that model rather than to the person who ran this experiment. It is recorded as a decision to be taken. Silently improving a frozen artifact is how a project loses the ability to say what it actually measured.

9. The honest limit - say it before they find it

THE ARCHITECTURAL GAP, STATED RATHER THAN SOFTENED

The encoder has **no way to know a pixel is missing**. The data loader replaces every missing value with the channel average, so 'I have no idea' and 'ordinary sea' arrive at the network as the same number. The code even computes a companion mask marking which values were real - and then discards it. So this is not graceful degradation; it is degradation while being told nothing. A test parses the source and **fails** if that mask ever starts being returned, at which point this page's framing is out of date and must be rewritten rather than relaxed.

IN ONE LINE

Set out to measure robustness to monsoon cloud and instead measured a warm bias in the deliverable, with the convenient reading explicitly rejected.

0.8950

DEGC AT 15% BLANKED - BETTER, AND NOT ROBUSTNESS

60x

THE NOISE FLOOR, SO THE DIP IS REAL

+0.50 degC

THE REAL DEGRADATION AT TOTAL BLINDNESS

Feature 15 of 17. Next: Feature 16.